

HV BATTERY SYSTEM

Temperature Limits

Battery pack and vehicle can operate safely in limits from 0°C to 50°C.

NOTE

To control the battery temperature of the high voltage battery the air conditioner is used to cool down the battery and may switch on automatically without request from control panel which may generate noise from operation of the air conditioner compressor and cooling fan.

Hv Battery Life & Maintenance

This Vehicle comes with a battery life of eight years or 160,000km warranty for the battery, whichever is earlier. Regular service of the vehicle and charging protocol to be followed to maximize the battery life.

Energy Information

The vehicle battery pack has a maximum energy of 40.5kWh. Energy retention capacity deteriorates over several cycles of

usage and hence range deterioration happens overtime.

This decrease in range during the end of life of battery is expected and is not considered as a malfunction of the battery pack. During these conditions, it is recommended to contact an authorized service center for inspection.

Brake Energy Recovery System

The vehicle features energy regeneration system, which regenerates expended energy during coasting or braking during the drive. This system allows the battery to be recharged under the above mentioned conditions.

Please note, the regeneration system does not fully recharge the battery, it only provides a chance to recover a portion of energy that would be lost during braking. When you release the accelerator or press the brake pedal, energy flows from wheels to high voltage battery, thereby charging it.

Regeneration is done by converting driving force (kinetic energy) into electrical energy that is stored in the Li-ion battery while the vehicle is decelerating or being driven downhill. This is called regenerative braking

Heavily Discharged High Voltage System

In the case of a heavily discharged HV battery, there is a chance that the low voltage battery is discharged as well. In this case, please contact your nearest TATA MOTORS EV service centre for further assistance. Do not try to jump start the vehicle or tow the vehicle without guidance from the service assistant.

Heated High Voltage Battery

In such a condition, the battery has safety logics to limit the performance or disconnect by itself with prior warning. The vehicle should be stopped and allowed to cool down and TATA MOTORS EV service centre must be contacted for rectification.

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Long Storage Of Vehicle With Respect To Hv Battery Pack

The HV battery undergoes discharge at a rate of approximately 2% over a period of 30 days in storage. Do not allow the vehicle to be discharged to 0% in storage. It is recommended that the vehicle must be charged to a charge level in between 30% to 50% before leaving the vehicle for long time storage. After this time period the vehicle must be charged to 100% using Normal Charging before use.

High Voltage System Failure

In the case of high voltage system failure, which may arise due to various reasons, contact TATA MOTORS EV service centre for further assistance.

Predicting Energy Usage

The vehicle battery energy usage is displayed in the instrument cluster in the form of estimated range. This range is updated by the system algorithm, depending on the driving conditions.

Displayed range in the instrument cluster is a tentative number based on drive route,

driving pattern and usage pattern history over the past drives. It is recommended keep a 20km buffer in estimated range before planning the trip

Disposal

The disposal of an HV Battery must be done with utmost care and will be carried out by TATA MOTORS after sales service at the end of the battery life time or if the battery pack has passed its warranty period.

NOTE

- It is advised to contact TPEM authorised service centres which shall guide & help in dismounting, handling and disposal through agencies who are certified by central/state pollution control board & obtain certificate of disposal from these agencies as proof of sustainable disposal.*
- If you decide not to use the recommended TATA Motors Authorised EV service center or TATA Motors Authorised EV Workshop to dispose*

of your high voltage battery, the responsibility of the consequences of environmental pollution or accidents must be borne solely by you, the owner of the vehicle.

- Customers who wish to dispose of battery by themselves shall deal only with registered entities (list of these is available on the CPCB/SPCB website) after duly verifying validity of necessary registration documents. After disposal the EPR certificate has to be obtained by the customer from the entity. It is request that customer shall provide this certificate to nearest TPEM service centre which is to be kept for records & submitted to central pollution control board as proof of disposal according to policy guidelines.*

WARNING

EV battery contains materials like Lithium Iron, graphite, plastic & steel etc

which can have impact on environment and are harmful if not handled/disposed of carefully. There is a risk of severe burns and electrical shock that may result in serious injury while additionally posing a risk of environmental damage.

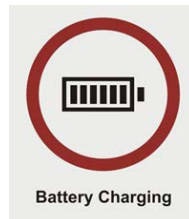
TIPS TO CONSERVE BATTERY LIFE

1. Battery Charging

It is advisable to charge the vehicle upto 100%. This ensures accurate SoC calculation and consistent energy content intake during charging. Charging to 100% also maintains battery health for longer time and ensures better range throughout the life cycle of the vehicle.

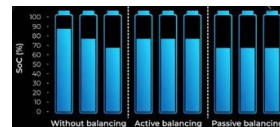
It is also advisable to slow or home charge the vehicle to 100% as much as possible. Slow Charging and full charging is the best way to charge your vehicle to maintain the health of the high voltage battery.

If fast charging is used, it is advisable to slow or home charge the vehicle from less than 20% charge to 100% for every 4 fast charging cycles.



2. Cell Balancing Or Equalization

Cell balancing or equalization and SoC calibration happens during charging, especially at higher SoC, > 90%. This causes charging to take longer time above 90%. Allow the vehicle to reach 100% SoC before terminating the charging session, as both cell balancing and SoC calibration are essential steps for any battery to perform optimally for a prolonged time.



3. Extreme Temperatures

Extreme temperatures impact upon how well a battery can maintain its level of charge. Avoid charging the high voltage battery when it is hot and vehicle is driven for long distances especially in summers. Allow vehicle temperature to cool down before charging. It is always



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better to charge EV when ambient temperature is low especially at night.

4. Vehicle Parking

Avoid parking vehicle in direct sunlight for long duration. Try to park it under shade, tree etc. This helps in keeping battery temperature low resulting enhance battery life over the life cycle.

